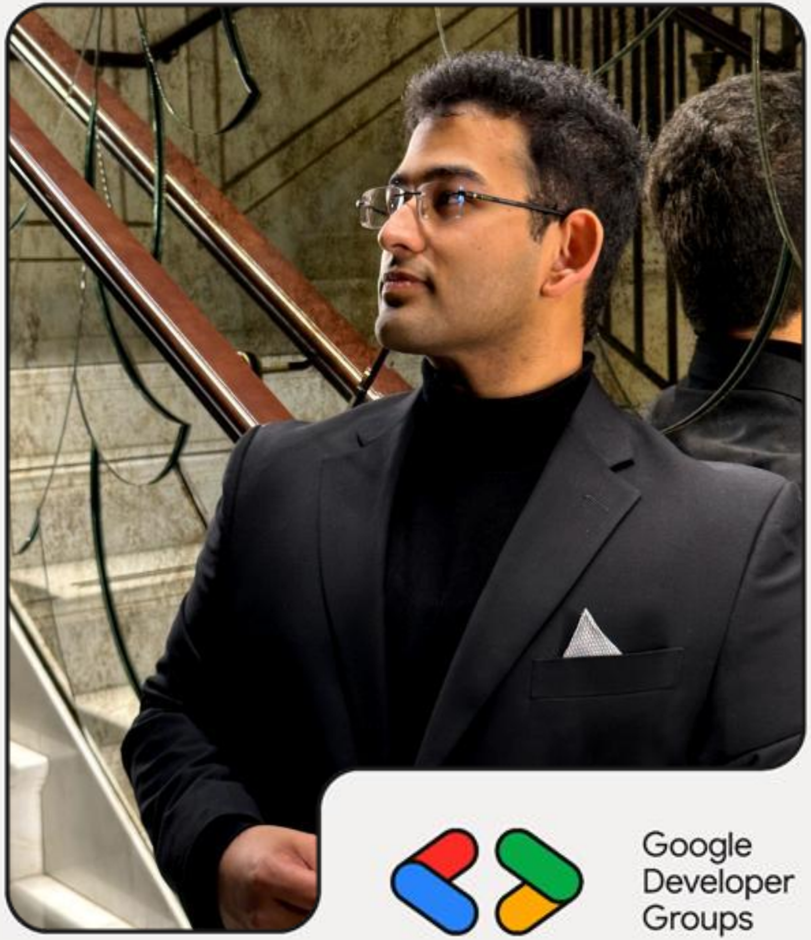
Build with AI

Dubai, UAE

Building And Deploying An Agentic AI App With Google AI Studio And ADK

Rohan Mitra

Google Developer Expert – Cloud AI
ML Engineer @ Bayut | Dubizzle
Research Assistant for ML @ AUS

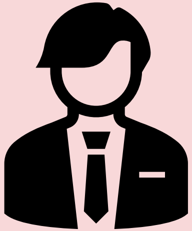


Google Developer Groups

LLMs vs Agentic AI

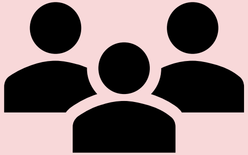
LLM

- Reasoning and knowledge, but cant take actions
- Stateless
- Passive



Agent

- Body = tools
- Job Desc = instructions & goals
- Autonomous
- Stateful
- Take actions till job is done



Multi-Agent

- Talented and focused individuals
- Get tasks to help achieve goal
- Focused
- Modular, reliable, debug-able

Why ADK?

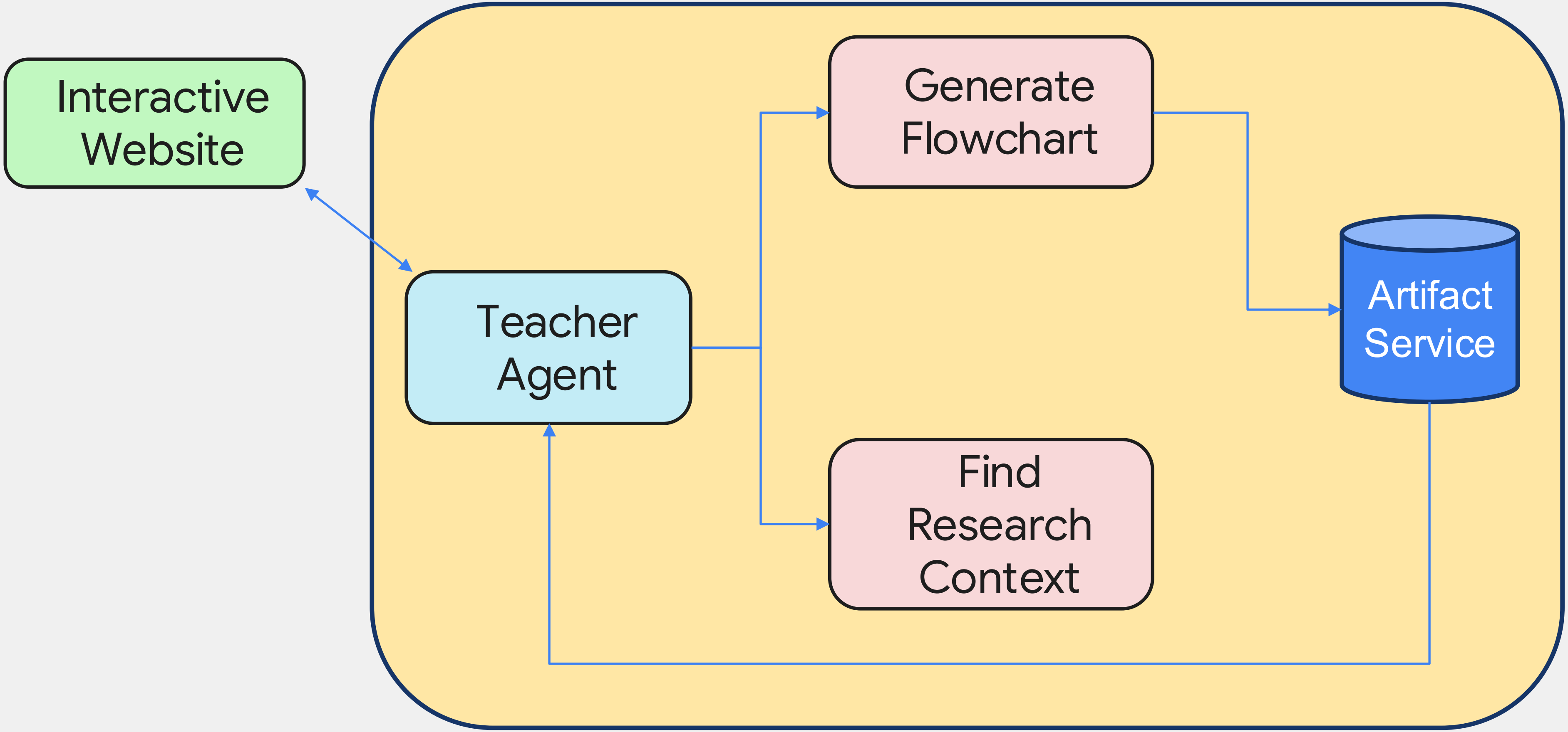
- Need control to build complex agents and workflows
  - Modularity
  - Reusability
  - State management
- Easy deployment and integration
- Quick set up while being scalable

Case Study – Research Explainer

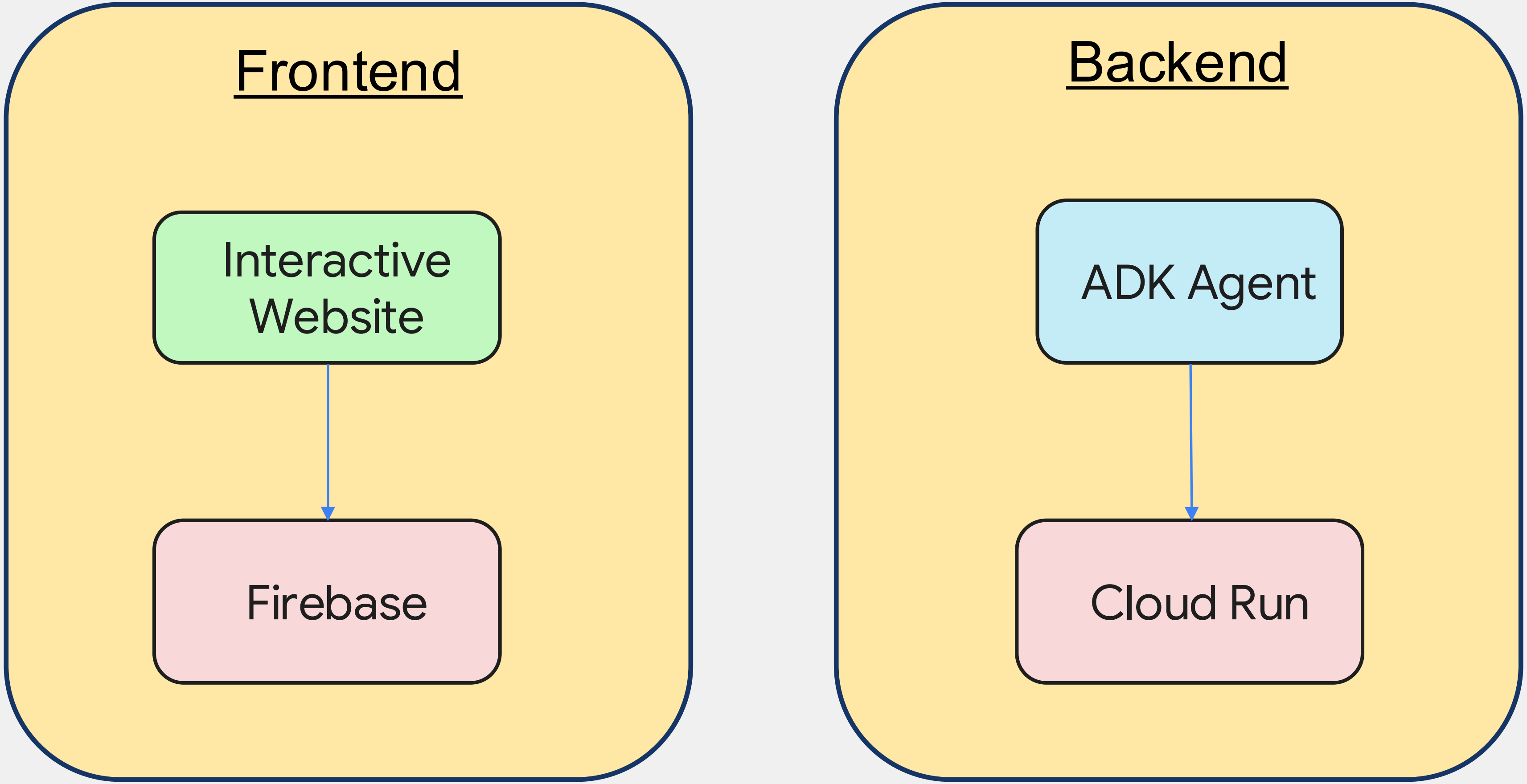
- Take a research paper / technical document and explain it while generating required diagrams and charts

- One Agent to read and understand the paper
- Tool call to generate flow charts – show processes
- Tool call to find external research context for a concept discussed – show applications

Our Architecture



Our Deployment Strategy



Artifact & Sessions

Artifacts

- Any large object that is loaded/created
- Includes files, images, data, etc
- **Artifact service:** A temporary storage space for the session
- Easy way to transfer large objects between agents
- Saves to RAM / GCS / Redis

Sessions

- Agent's memory of this conversation
- Handled by ADK
- **Session service:** Where the session details will be saved (temp)
- Important when multiple agent pods exist – share the same memory
- Saves to RAM / GCS / Redis

Build with AI
Dubai, UAE

Demo
Code on Github

github.com/ro1406/research-paper-explainer

Google Developer Groups

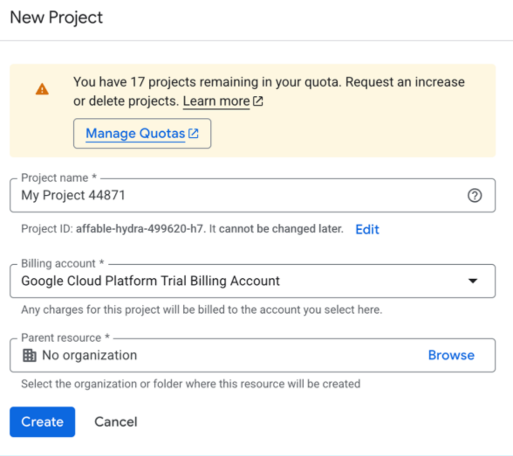
Deployment Options

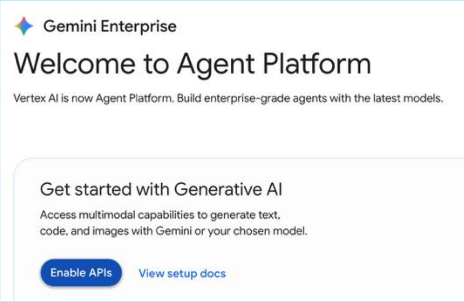
- Deploy directly to GCP – Cloud Run
- Non-GCP deployments – Docker container
  - Single service
  - Host as `adk api_server –port 8000`
- Identify good Artifact services
  - RAM – local / low traffic / single deployment
  - GCS good for GCP deployments
  - Redis sidecar for kubernetes deployments
- Serverless deployments
  - Event driven
  - Scalable (to 0 if needed)

Medium.com Article

Deployment - Setup

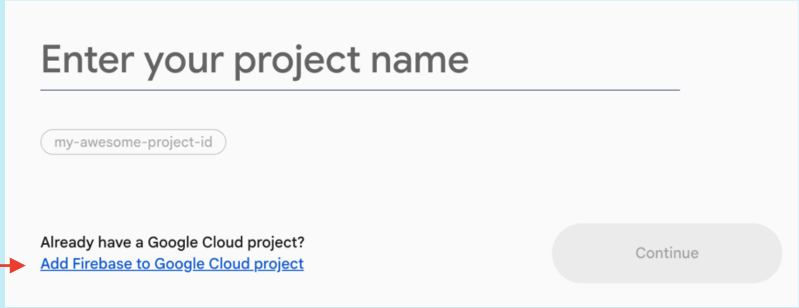
- Install GCloud: `https://docs.cloud.google.com/sdk/docs/install-sdk`
- Install Firebase CLI: `npm install -g firebase-tools`
- Create GCP Project
- Enable VertexAI API (Skip if you're using AI Studio API Key)





Deployment – Link Firebase

- Go to `console.firebase.google.com`
- Click "**Add Project**" -> Choose "**Add Firebase to a Google Cloud Project**"
- Select existing GCP Project
- Follow the prompts



Deployment – Local GCP Setup

- In local console, run gcloud auth login
- Run gcloud init and set project name
- Export the required env vars
  - `GOOGLE_CLOUD_LOCATION`
  - `GOOGLE_GENAI_USE_VERTEXAI` (if using VertexAI API, else `GOOGLE_API_KEY`)
  - `GOOGLE_CLOUD_PROJECT_ID`
- Set project name in `.firebaserc`
- Run `firebase login`

Deployment – Cloud Run Deploy

```
gcloud run deploy research-explainer \
  --source . \
  --region us-central1 \
  --min-instances=1 \
  --max-instances=1 \
  --allow-unauthenticated \
  --set-env-vars="GOOGLE_API_KEY=<your_api_key>"
```

Deployment – Firebase Deploy

- Set the URL from cloud run deploy into `index.html BACKEND_URL`
- `firebase deploy --only hosting`

Build with AI
Dubai, UAE

Thank You!

rohanmitra.dev

Google Developer Groups